Discourse Heuristics For Paradoxically Moral Self-Correction

Anonymous ACL submission

1 More Experimental Results

In this document, we present additional experimental results on several models, including Phi-3.5-mini-instruct (3.8B), Llama-3-8B-Instruct, and Mistral-7B-Instruct-v0.3. These results extend our analysis across diverse model architectures and parameter scales. Overall, the findings align with the conclusions reported in our main paper.

1.1 Results of Phi-3.8b

Table 1 presents the self-correction performance of Phi-3.5-mini-instruct after fine-tuning on three discourse constructions. All constructions demonstrate performance gains over the baseline, validating the effectiveness of our approach. Consistent with the findings in the paper, the discourse without the `Statement` component achieves the highest improvement, while removing `Context` results in a performance drop compared to the full construction (All), underscoring the critical role of contextual information.

Table 2 summarizes in-domain and out-of-domain generalization experiments for Phi-3.5-mini-instruct (3.8B). The results mirror those reported for LLaMA-3-2B (3B) in the paper, further demonstrate the robust out-of-domain generalization capabilities of LLMs of such sizes.

1.2 Results of larger models

This section presents experiments on larger models, LLaMA-3-8B-Instruct and Mistral-7B-Instruct-v0.3, using LoRA fine-tuning with a rank of 64 and a learning rate of 1e-5. It should be noted that we select Llama3-8b due to its architectural consistency with the 1B/3B models in our study. However, since the BBQ benchmark is widely used, this model has already been fine-tuned on it and achieves near perfect performance. Nevertheless, our framework still demonstrates measurable improvements.

Table 3 summarizes the self-correction performance across all discourse constructions. Compared to smaller models, the 7B/8B parameter models achieve superior performance, with LLaMA-3-8B-Instruct attaining perfect accuracy in four out of five tasks. These results further validate the effectiveness of our proposed constructions for larger-scale LLMs.

Compared with smaller LLMs, there are some important observations. Removing the `Context` or `Action` components do harm to the performance relative to the full construction (`All`). However, only three out of ten **All - Statement** (specifically, *Disability* in Llama, *Age* and *Nation* in Mistral) underperform the **All** construction. This suggests that that explicit stereotype awareness is *not* essential for successful self-correction. These findings align with the shallow heuristics hypothesis presented in Section 4.2 of our paper.

Table 4 present self-diagnosis performance when self-correction is enhanced. There are two key-observations: **(1)** The overall performance is not higher than smaller models significantly (primarily between 0.6-0.7). **(2)** The conflicts are slightly mitigated in larger-scale models. The empirical findings necessitate fine-grained analysis of self-diagnosis, which is essential to understand how LLMs perform moral tasks.

Table 5 and Table 6 demonstrate the in-domain and out-of-domain generalization capability of both models. The results show strong generalization of moral self-correction in LLMs for larger-scale models, thereby extending the finding from section 5.2.

Phi3.5-mini	Age	Nation	Gender	SES	Disability
Baseline	.9176	1	.9492	.9985	.9803
All	.9574	1	.9637	.9985	.9934
All - Context	.9347	1	.9528	.9985	.9868
All - Statement	**.9716**	**1**	**.9819**	**.9985**	**1**

Table 1: Experimental Results on Phi-3.5-mini-instruct for **Self-correction** Across Different Discourse Constructions. **All** includes all possible components: context, statement, action. **All - *** indicates a component was removed from the setting of **All**. The optimal performance is highlighted in **bold**. Across all experiments, **All - Statement** contributes to the optimal performance in nine out of ten experiments, suggesting that LLMs do not need stereotype awareness for successful self-correction.

Phi3.5-mini	Age	Nation	Gender	SES	SexOrientation	Disability	Physical	Religion
Baseline	0.9176	1	0.9492	0.9985	0.9803	0.995	0.970	0.943
Mixed	0.9716	1	0.9691	0.9985	0.9934	0.9954	0.9824	0.9683

Table 2: Self-correction In-domain and Out-of-domain Generalization of Phi-3.5-mini-instruct By Mixing the Fine-tuning Corpus of 5 Representative Stereotypes. Phi-3.5-mini-instruct(3.8b) shows both good in-domain and out-of-domain generalization

Llama3|Mistral	Age	Nation	Gender	SES	Disability	Age	Nation	Gender	SES	Disability
Baseline	.906	.987	.955	.897	.993	.838	.837	.695	.810	.849
All	.974	1.0	1.0	.964	**1.0**	.878	**.887**	.871	.955	.882
All - Context	.926	.983	.982	.941	1.0	.869	.880	.742	.821	.862
All - Statement	.983	**1.0**	**1.0**	**1.0**	.938	**.938**	.857	.860	**.988**	.882
All - Action	.957	.997	1.0	958	1.0	.847	.833	.790	.929	.875
Action	.940	.990	.969	.927	1.0	.889	.843	.735	.830	.862
All-Statement-subaction	**.986**	.993	1.0	.973	1.0	.898	.873	**.877**	.939	.875

Table 3: Experimental Results on Llama3-8b-it/Mistral-v0.3-7b-it for **Self-correction** Across Different Discourse Constructions. **All** includes all possible components: context, statement, action. **All - *** indicates a component was removed from the setting of **All**. The optimal performance is highlighted in **bold**. Across all experiments, only three out of ten **All - Statement** (**Disability** in Llama, **Age** and **Nation** in Mistral) do harm to the performance compared to the **All**, suggesting that LLMs do not need stereotype awareness for successful self-correction. Note: We select Llama3-8b due to its architectural consistency with the 1B/3B models in our paper. However, since the BBQ benchmark is widely used, this model has already been fine-tuned on it and achieves near perfect performance. Nevertheless, our framework still demonstrates measurable improvements.

Llama3|Mistral	Age	Nation	Gender	SES	Disability	Age	Nation	Gender	SES	Disability
selfdiag baseline	.665	.540	.604	.777	.697	.645	.550	.682	.635	.618
selfcorr→ selfdiag	<u>.713</u>	<u>.633</u>	<u>.655</u>	<u>.882</u>	<u>.697</u>	<u>.656</u>	<u>.560</u>	.603	.600	<u>.632</u>

Table 4: Experimental Results of Llama3-8b-it/Mistral-v0.3-7b-it for Test Performance in Self-diagnosis Capability While Improving Self-correction. For all ten experiments, there are conflicts for only two of them.

Llama3|Mistral	Age	Nation	Gender	SES	Disable	Age	Nation	Gender	SES	Disable
Baseline	.906	.987	.955	.897	.993	.838	.837	.695	.810	.849
Individual	.983	1.0	1.0	1.0	.938	.938	.857	.860	.988	.882
Mixed	.992	.997	.996	.969	.993	.966	.980	.964	1.0	.967

Table 5: In-domain Generalization of Llama3-8b-it/Mistral-v0.3-7b-it By Mixing the Fine-tuning Corpus of Five Representative Stereotypes. `Individual` represents fine-tuning with the discourse within one stereotype in Section **??**. `Mixed` means that we mix the fine-tuning dataset of five stereotypes and test the fine-tuned model on each stereotype. The 3B model shows good in-domain generalization but 1B model does not, implying the generalization of heuristics does rely on model sizes.

Llama3|Mistral	**SexOrientation**	**Physical**	**Religion**	**SexOrientation**	**Physical**	**Religion**
Baseline	.947	.959	.920	.759	.849	.785
Mixed	.988	.989	.983	.951	.972	983

Table 6: Out-of-domain Generalization of Llama3-8b-it/Mistral-v0.3-7b-it By Mixing the Fine-tuning Corpus of 5 Representative Stereotypes. 7B/8B models show good out-of-domain generalization